

FMR OPERATIONS v3

Production Rollout & Operations Guide

Launch baseline: FMRCore 3.0.0-alpha.30.4 | Library v35

Release decision	CONTROLLED GO
Git baseline	5b2e12d5f204b35a00452851342d4d2af8103d37
Verified migration	Alpha 30.4 fractional Size repair verified in a new migration
Audience	Leadership, System Owners, Material Admins, Field Users

Prepared for controlled production deployment - August 12, 2026

This guide reflects the verified GitHub release and the current TEST database review.

1. Executive Production Readiness Decision

CONTROLLED GO

The application code and migration/parser path are ready for production use. The remaining historical FMR backlog is not a launch blocker. The hard launch condition is that production must use a clean PRODUCTION data environment rather than simply relabeling the current TEST database.

The current release has passed the most important launch checks: source-control parity, Alpha 30.4 fractional Size correction, evidence-backed reconciliation of previously imported Size defects, a new post-fix migration with correct Size values, operational integrity diagnostics, backups, audit controls, and functional Field/Admin/Owner workflows.

Leadership can proceed with deployment for the approximately 20 intended users. The expected concurrency is comfortably below current Apps Script simultaneous-execution limits; the more relevant scaling concern is Google Sheets read/write latency, which the application already mitigates through indexed reads, caching, batching, and lock-protected writes.

Readiness scorecard

Gate	Status	Evidence / action
GitHub release baseline	PASS	Alpha 30.4 commit is the latest main-branch baseline.
Core / Bound dependency	PASS	FMRCore 3.0.0-alpha.30.4; immutable library version 35.
Fractional Size parser	PASS	New imports restore 3/4, 3/8 and early numeric Sizes correctly.
Existing Size reconciliation	PASS	Evidence-backed reconciliation applied with automatic backup and audit log.
Known bad Size patterns	PASS	No 4/3, 2/1, 8/3, or 8/5 remain in current Bulk_Import_Lines or published FMR_Line_Items.
FMR 221 spot check	PASS	Published values now show 3/4, 3/4, 1/2, then 3/4 on lines 4-7.
TEST integrity / schema / controls	PASS	Latest health run shows all three as PASS.
TEST overall health	WARN	Only stale test Backorder and Bag fixtures remain; do not carry them into PROD.
Production data environment	REQUIRED	Create a clean PROD database with a unique fingerprint before user cutover.
Remaining historical backlog	DEFERRED	May be imported after go-live in controlled off-peak batches.

2. What Was Verified Before This Recommendation

Release and source control

- Latest GitHub baseline is commit 5b2e12d5f204b35a00452851342d4d2af8103d37 (Alpha 30.4 size parser corrections).
- Core runtime version is 3.0.0-alpha.30.4.
- Bound is pinned to immutable FMRCore library version 35.
- The bulk-import parser fingerprint is ALPHA30_4_SIZE_ORIENTATION_REPAIR_V1 and the parser calls the Alpha 30.4 Size-normalization helper.
- The size reconciliation service uses Owner authorization, ScriptLock, stale-preview protection, automatic pre-change backup, per-cell validation, detailed reconciliation logging, audit logging, and a post-apply verification pass.

Database and migration evidence

- The current TEST database had at least 373 published FMRs at the latest scheduled health check, before the final Alpha 30.4 migration validation.
- The Alpha 30.4 reconciliation created backup BACKUP-3547A9C2-C28A-4879-AE7A-11F69B0706E4 and reconciled 157 evidence-backed candidates across 324 cell patches.
- The reconciliation log confirms the former reversed values were corrected, including 4/3 -> 3/4, 2/1 -> 1/2, 8/3 -> 3/8, and 8/5 -> 5/8.
- The latest 5-file migration discovered 20 FMRs and completed all 5 files. Eighteen were published/already existing and two were blocked because the same Official FMR Number (20) appears twice in one source workbook - a legitimate source-data block, not a parser defect.
- The latest Alpha 30.4 import explicitly recorded a converted March 8 date being restored to 3/8, March 4 to 3/4, and an early spreadsheet date to numeric Size 1. This is direct evidence that the corrected parser is active.

Important distinction

A migration job can finish COMPLETE_WITH_EXCEPTIONS and still be healthy. Genuine source problems such as duplicate FMR numbers, missing FMR numbers, or blank requested quantities should remain blocked. Do not weaken the validator just to make the migration summary green.

3. Application Overview

FMR Operations v3 is a Google Apps Script web application backed by an indexed Google Sheets data store. It replaces a spreadsheet-only workflow with controlled role-based interfaces, material state tracking, immutable operational transactions, audit history, backup/recovery controls, and a governed historical migration process.

Primary user roles

Role	Primary interface	Writes?	Typical responsibilities
Field User	Search	Yes	Search by FMR/ISO, confirm material located, Bag & Tag, issue material, request backorder.
Material Admin	Admin	Yes	Review operational queues, decide Backorders, use Admin register/dashboard.
System Owner	Owner	Yes	All operational access plus migration, user maintenance, diagnostics, backups, controlled corrections and recovery.

Core records

Data area	Purpose
FMR_Header / FMR_Line_Items	Published request master data and current material-state totals.
Search_Index / Operational_Index	Indexed lookup and operational queue acceleration.
Material_Transactions	Append-only operational ledger for confirmations, issues, Bags and Backorder actions.
Bag_Tag_Header / Bag_Tag_Items	Reserved/bagged material and issuance from Bags.
Backorder_Requests	Pending, confirmed, rejected, returned and fulfilled Backorder workflow.
Field_Notifications	Field-facing status/decision notices.
Audit_Log	System audit trail for controlled changes and exceptional actions.

Data area	Purpose
Owner_Corrections	Owner-governed transaction reversal/correction records.
Historical_Migration_* / Bulk_Import_*	Historical source inventory, parsing, staging, exceptions and publication.
Size_Reconciliation_Log	Permanent evidence of the Alpha 30.4 Size repair applied to earlier imports.

4. Data Integrity and Governance Principles

- Requested quantity does not mean material has been physically received. Availability is established through operational confirmation.
- Material transactions are treated as immutable history. Owner corrections append compensating REVERSAL_* transactions instead of editing or deleting original transaction rows.
- Bagged material is reserved and cannot be treated as generally available unless released or reassigned through authorized workflow.
- Backorder actions follow explicit state transitions and preserve the request/decision history.
- Write-sensitive functions use locks and validation to reduce concurrent-write collisions.
- Exceptional Owner actions create backups before mutation and leave an Audit_Log trail.
- Historical migration is REQUEST_ONLY: legacy Issued, Back Ordered and Action Taken values remain source evidence and do not manufacture live Material_Transactions.
- Warnings can be explicitly approved for publication; hard validation errors remain blocked.

Do not edit the backing spreadsheet as an operational shortcut

Production users should use the web application. Direct spreadsheet edits can bypass validation, index invalidation, transaction logging, locks, and audit controls. Spreadsheet access should be governed as an administrative dependency, not used as the normal user interface.

5. Required Production Topology

Do not promote the current TEST database by changing its label.

The TEST database contains acceptance fixtures, test transactions, Bags, Backorders, reversals/corrections, migration history, and known stale test records. Production should have its own database fingerprint and operational history from day one.

Component	Production requirement
Core library	Use the frozen Alpha 30.4 source and immutable FMRCore library version 35.
Bound web app	Create/maintain a versioned PROD deployment. Do not hand users the /dev URL.
Production database	Separate clean Google Sheet using the same v3 schema. ENVIRONMENT_NAME=PRODUCTION.
Transaction mode	Start READ_ONLY for final smoke validation; switch to ENABLED only at formal go-live.
Backup folder	Dedicated accessible PROD backup folder; verify every System Owner can create a backup.
Users	Load only the approved production roster and least-privilege roles.
Historical FMRs	Load only what is needed at launch; remaining history may be migrated after production.
Operational tables	Start without TEST transactions, Bags, Backorders, Field

Component	Production requirement
	Notifications, Owner Corrections, or stale fixtures.
Health / audit / backup history	Start new PROD histories rather than copying TEST operational history.

6. Production Rollout Plan

Phase 0 - Freeze the release (15 minutes)

1. Freeze feature development at GitHub commit 5b2e12d5f204b35a00452851342d4d2af8103d37. Only release-blocking fixes are allowed until after cutover.
2. Create a release tag or protected release branch for the production baseline (recommended naming: prod-2026-08-12 or release/3.0.0-alpha.30.4).
3. Record the Bound deployment version, FMRCore library version 35, and the production database ID in the cutover record.

Phase 1 - Provision clean PROD (30-60 minutes)

4. Create the production database from the approved v3 schema/bootstrap process. Do not use the current TEST spreadsheet as the live production database.
5. Set ENVIRONMENT_NAME to PRODUCTION and TRANSACTION_MODE to READ_ONLY during provisioning.
6. Configure the dedicated PROD backup folder and retention settings.
7. Load the approved production user roster. Keep System Owner access to the minimum needed for continuity.
8. Load/import the validated FMR master data required for day-one operations. It is acceptable to leave the rest of the historical backlog for later.
9. Confirm Material_Transactions, Bag_Tag_Header, Bag_Tag_Items, Backorder_Requests, Field_Notifications and Owner_Corrections contain no copied TEST operational activity.
10. Run the storage/integrity/health diagnostics and create a baseline PROD backup.

Phase 2 - Controlled smoke test in READ_ONLY / commissioning mode (30-45 minutes)

11. Open the PROD URL as a System Owner, a Material Admin and 2-3 representative Field Users.
12. Confirm every account completes OAuth authorization and lands in the correct role/interface.
13. Validate FMR search and ISO+sheet search against known production FMRs.
14. Confirm Admin register/dashboard load and the Owner readiness view shows the PROD environment/fingerprint.
15. Use one designated commissioning/test FMR if a write test is required. Do not use a real material request for destructive reversal testing.
16. Verify one controlled end-to-end path appropriate for commissioning: locate/confirm -> Bag or issue -> Backorder request/decision, as practical.
17. Confirm Audit_Log, Material_Transactions and operational queues reconcile after the smoke test.
18. Create another PROD backup after commissioning if any writes occurred.

Phase 3 - Go live

19. Set TRANSACTION_MODE to ENABLED and clear any maintenance message.
20. Distribute only the versioned PROD web-app URL, not the TEST or /dev URL.
21. Open access in waves: System Owners and Material Admins first, then Field Users.
22. Keep the first 30-60 minutes focused on normal search and a small number of real transactions; avoid mass historical migration during initial cutover.
23. Run Operational Readiness after the first several transactions and again near the end of the first shift.

Phase 4 - First-day stabilization

24. Review the Audit Log, failed executions, stale queues, and user authorization issues.
25. Confirm scheduled/manual backups are succeeding in the PROD backup folder.
26. Collect user-reported search or transaction latency and distinguish one-off Apps Script variability from repeatable slow paths.
27. Import additional historical FMRs only after the live workflow is stable.

7. User Onboarding and Access

The current Bound manifest runs the web application as USER_ACCESSING. In practical terms, each person uses the app under their own Google identity and must authorize the required Google scopes. The application should verify authorization before sensitive Drive/Sheets calls.

Role	Recommended production permissions
Field User	Can_Search=YES; Can_Field_Transact=YES; Admin/Owner permissions=NO.
Material Admin	Can_Search=YES; Can_Admin_Backorder=YES; Owner permission=NO. Enable Field transact only if job duties require it.
System Owner	All required permissions. Keep the group small and explicitly verify access to the PROD database and PROD backup folder.

First-login instructions

- Use the approved company Google account assigned to the production Users table.
- Open the PROD URL and complete the OAuth permission screen. Do not selectively deny a scope required by the app.
- If authorization fails, stop and contact the System Owner rather than repeatedly retrying transactions.
- Confirm the header shows PRODUCTION and the expected FMRCore version before entering real operational transactions.
- Bookmark the PROD URL only after successful login; remove old TEST bookmarks from field iPads if practical.

Owner permission check

A real TEST failure already demonstrated that a System Owner without access to the configured backup folder cannot complete a correction that requires an automatic backup. Verify backup-folder access for every production Owner before go-live.

8. Field User Quick Start

Task	What to do	Key rule
Find material	Search by FMR number or ISO/line + sheet.	Do not assume requested quantity means received/available.
Confirm available	Use Confirm Available only for material physically located/confirmed.	Located quantity cannot exceed the remaining requirement.
Bag & Tag	Reserve located material to a Bag/Tag and storage location.	Bagged material is reserved.
Issue material	Issue directly/from available/from Bag as appropriate.	Capture the recipient name; transaction history is retained.
Request Backorder	Create a Backorder request for not-yet-located need.	Admin decision remains a separate controlled step.
Returned/Rejected notice	Review the field notice and follow the displayed status.	Do not bypass the workflow with direct spreadsheet edits.

9. Material Admin and System Owner Quick Start

Material Admin

- Use the operational dashboard/register for queue awareness and exact FMR lookup.
- Review Pending Backorders and select Confirm, Reject or Return based on material-control disposition.
- Use notes to preserve decision context. Do not edit Backorder_Requests directly in Sheets.
- If a queue appears stale, verify the underlying FMR and transaction history before treating it as a data defect.

System Owner

- Own the production user roster, configuration, migration controls, backups/recovery, diagnostics and exceptional correction workflows.
- Use Owner correction/reversal only when a legitimate transaction must be undone. Original Material_Transactions are never deleted or edited.
- Create/verify a backup before significant maintenance. Exceptional correction and Size reconciliation already enforce this automatically.
- When a historical migration is interrupted, Resume the same RUNNING job. Do not create a second job against the same folder.
- Use Abandon only to close an interrupted migration you intentionally do not want to resume; abandonment does not delete FMRs already published.

10. Historical Migration After Go-Live

Leadership has accepted that the remaining historical FMR backlog can be migrated after production. This is a reasonable deployment decision because historical publication is separated from live material transactions by the REQUEST_ONLY policy.

- Prefer off-peak migration windows so Excel conversion/parsing does not compete with active field operations.
- Use a dedicated source folder and Preview Folder before starting.
- Keep generated migration working folders excluded from recursive discovery; Alpha 30.3+ already protects this path.
- Use Continue automatically and keep the Owner tab open. If the browser loop stops, reopen/resume the same job.
- Auto-publish only warning categories that have been explicitly accepted. Hard errors must remain blocked.
- Expected source-data blockers include duplicate FMR numbers, missing official FMR numbers, and blank requested quantities.
- Legacy Issued / Back Ordered / Action Taken values remain source evidence only and do not create live transaction history.
- After each meaningful batch, run Operational Readiness and a storage profile. Spot-check representative newly published FMRs in Field/Admin.

11. Performance and Expected Scale

For approximately 20 users with low probability of simultaneous heavy writes, current platform concurrency limits are not the immediate constraint. Google currently documents 30 simultaneous Apps Script executions per user and 1,000 simultaneous executions per script, with a 6-minute execution limit per script invocation. These quotas can change, so they should be treated as platform ceilings rather than an application service-level guarantee.

The practical performance risk is Google Sheets service latency. The project already reduced the most significant N-per-line read patterns using batched lookups, indexed search/operational tables, and ScriptCache. Warm acceptance measurements before the final migration phase were approximately 3.4 seconds for Field Search, 2.8 seconds for Owner Ledger, 0.5 seconds for Admin Dashboard, and 0.2 seconds for Admin exact-FMR search.

Launch performance expectation

A field search taking a few seconds is acceptable for this initial 20-user deployment. Investigate repeated regressions, not an isolated slow Apps Script invocation. If sustained usage grows substantially, move performance analysis toward measured service-call counts and data-store scale rather than premature architectural changes.

12. Production Monitoring Checklist

When	Check	Escalate when
Before go-live	PRODUCTION label/fingerprint, READ_ONLY, user roles, baseline backup, integrity/schema/control PASS.	Any environment mismatch, missing backup, or integrity/schema error.
First 30 min	Search latency, authorization failures, first transactions, Admin queues.	Repeated transaction errors, duplicate state, inability to authorize.
First 2 hours	Operational Readiness, transaction/audit reconciliation, backup accessibility.	Integrity error, unexplained negative/over-requested state, failed required backup.

When	Check	Escalate when
End of shift	Stale queues, rejected/returned notices, recent transactions, user feedback.	Stale warnings caused by live workflow rather than known fixtures.
Daily	Latest successful PROD backup age, health log, failed executions, user changes.	Backup exceeds configured age threshold or production warning cannot be explained.

13. Incident Response and Rollback

If a production defect affects transactions

28. Set TRANSACTION_MODE to READ_ONLY immediately and publish a short maintenance message.
29. Preserve the current database. Do not delete Material_Transactions, Bags, Backorders, or Audit rows as an improvised fix.
30. Capture the affected FMR number(s), user, approximate timestamp, action attempted and visible error.
31. Run Operational Readiness and review the relevant transaction/audit history.
32. If the issue is code-related, roll the Bound deployment back to the last known-good version while keeping the immutable Core dependency explicit.
33. If the issue is data corruption, use the most recent successful PROD backup as the recovery source and perform a controlled reconcile/restore rather than ad-hoc edits.
34. Re-enable transactions only after the root cause and data state have been verified.

If historical migration fails

- Do not treat a blocked historical FMR as a production outage.
- Resume an interrupted RUNNING job rather than starting another job for the same folder.
- Leave genuine source-data exceptions blocked and document them for later source correction.
- If a migration defect is found, stop additional migration while normal Field/Admin operations remain available.

14. Known Limitations and Operational Boundaries

- The backing store is Google Sheets, not a transactional relational database. ScriptLock and validation reduce concurrent-write risk, but the platform does not provide full ACID transactions.
- Apps Script and Google Sheets latency can vary even when code is unchanged.
- The web app executes as the accessing user, so authorization and file permissions are part of the production dependency chain.
- The Bound project uses an Apps Script library. Libraries improve reuse/governance but can add some startup overhead in UI-heavy call patterns; current performance is acceptable at the intended scale.
- Historical Excel conversion can be slow because each source workbook must be converted/opened/parsed; migration should not be used as a benchmark for normal Field search speed.
- The Configuration row APP_VERSION currently reflects an older descriptive value and should be cleaned up in a later housekeeping change. Runtime release identification is governed by the Core version/deployment baseline.
- Any TEST stale Bag/Backorder warnings are not acceptable justification for carrying TEST operational rows into PROD.

15. Go-Live Sign-Off

Recommended launch decision

Approve production cutover once the clean PROD database, role roster, backup folder, PROD deployment and smoke test are complete. The remaining historical FMR backlog may follow after go-live.

Launch gate	Owner / evidence	Status
Release baseline frozen	Git commit + deployment version	_____
Clean PROD database created	PRODUCTION environment + unique fingerprint	_____
Production users loaded	Approved roster / least privilege	_____
PROD backup verified	Successful baseline backup ID	_____

Launch gate	Owner / evidence	Status
Integrity/schema/control diagnostics	All PASS	_____
Field/Admin/Owner smoke test	Representative accounts	_____
Transaction mode enabled	Formal go-live time	_____
First post-launch health run	Operational Readiness reviewed	_____

Go-live date/time: _____ **Approved by:** _____

PROD URL / deployment ID: _____

PROD database ID / fingerprint: _____

Appendix A - Verification Basis

This rollout recommendation was prepared from the live project artifacts reviewed on August 12, 2026, including the current GitHub main branch, the current FMR Operations TEST database, migration/reconciliation history, backup history, user/configuration tables, and current Google Apps Script platform documentation.

Source	Verified item
GitHub - jonaDaniM/FMRv3	Alpha 30.4 release commit, Core version, parser fingerprint, Bound library v35, Size normalization and reconciliation service.
FMR Operations TEST database	Current schema, users, health log, migration history, reconciliation log, backup history, FMR 221, and known-size-pattern searches.
Google Apps Script - Web Apps / Authorization	USER_ACCESSING execution identity and user OAuth authorization behavior.
Google Apps Script - Quotas	Current 6-minute execution limit, 30 simultaneous executions per user, 1,000 per script.
Google Apps Script - Best Practices	Batch service operations and caching as primary performance practices.

Release baseline recorded in this guide: commit 5b2e12d5f204b35a00452851342d4d2af8103d37; FMRCore 3.0.0-alpha.30.4; Bound library dependency version 35.